2.6.6 Toxicology Written Summary

PrP-lowering divalent siRNA 2439-exNA

Other names: 2439-exNA, 2439-s4, PrP-siRNA

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LIST OF ABBREVIATIONS

ASO	antisense oligonucleotide
CSF	cerebrospinal fluid
di-siRNA	divalent short interfering RNA
ED_{50}	median effective dose
exNA	extended nucleic acid
GLP	Good Laboratory Practices
ICH	International Conference on Harmonization
ICV	intracerebroventricular
IT	intrathecal
OECD	Organization for Economic Co-operation and Development
PrP	prion protein
siRNA	short interfering RNA
TK	toxicokinetic

2.6.6 TOXICOLOGY WRITTEN SUMMARY

2.6.6.1 Brief Summary

PrP lowering divalent siRNA 2439-exNA (hereafter referred to as 2439-exNA) is a fully chemically modified oligonucleotide drug candidate designed to degrade the mRNA encoding PrP and intended for the treatment of human prion disease.

The nonclinical testing program included single administration GLP toxicity studies in rats and dogs with 28 days post-dosing recovery period and safety pharmacology endpoints, in vitro genetic toxicology studies, primary pharmacology studies and an in silico analysis of secondary pharmacology targets (see Pharmacology Written Summary 2.6.2).

Rationale for choice of species

The rat and beagle dog were chosen as the toxicology species for 2439-exNA. Neither species is pharmacologically responsive to inhibition of PrP expression by 2439-exNA, therefore the studies only assessed off-target effects, such as those related to oligonucleotide chemistry. However, an overwhelming body of evidence indicates that partial or even complete PrP lowering will not have adverse effects. Decades of evidence from animal models as well as humans support the safety of PrP lowering[1]. PrP knockout mice were first engineered in 1992 and initially defied any efforts to identify a phenotype other than total resistance to prion disease[2]. Additional lines of PrP knockout mice using alternative gene disruption strategies were later engineered, for a total of 8 extant PrP knockout lines[3]. PrP knockout cattle and goats have also been engineered[4,5], and naturally occurring knockout goats with a premature termination codon were later identified as well[6]; all these knockout animals proved healthy. Postnatal knockout in neurons using a Cre recombinase system also proved well-tolerated in mice[7].

Because PrP is a highly studied gene, a large number of scientific papers have been published on different observations in PrP knockout mice, however, a large majority of assertions of PrP knockout phenotypes have never been replicated, and many have been refuted or ascribed to genetic confounders[8]. Following 18 years of intensive study, just one knockout phenotype emerged as reproducible[9]. Aged homozygous PrP knockout mice develop a mild peripheral neuropathy associated with insufficient maintenance of myelin on peripheral nerves[9]. This phenotype is not observed in the central nervous system, but only the peripheral nervous system; not in heterozygous knockout animals, but only in homozygote mice; and not in young but only in aged animals[9]. The phenotype has replicated in PrP knockout goats[10]. While the change is histologically evident when sciatic nerves are sectioned and stained, observing any overt phenotypic impact of this neuropathy has proven challenging. Delays in hot plate response and reduced grip strength were observed in aged homozygous knockout mice on a Balb/c genetic background but not on a C57BL/6 genetic background[9], likely simply reflecting the phenotypic subtlety of the phenotype and the difficulty of powering studies for these variable endpoints. The adhesion GPCR Adgrg6, expressed on Schwann cells, was identified as a receptor for an N-terminal peptide naturally cleaved from PrP; deficiency of this signaling peptide in PrP knockout mice apparently mediates the myelin maintenance phenotype[11]. Adgrg6 is not expressed in the CNS, providing an explanation for why the myelin phenotype is observed only in peripheral nerves.

Human genetics also supports the safety of PrP lowering. The advent of genomic databases of genetic variation in the general population, such as ExAC, gnomAD, and UK BioBank, has provided an opportunity to observe naturally occurring knockout alleles in the human population. N-terminal

premature nonsense or frameshift variants, expected to cause a total loss-of-function, are found in a heterozygous state in approximately 1 in every 25,000 individuals in these databases[12,13]. These individuals occur randomly across various cohorts ascertained for genetic sequencing studies, rather than being enriched for any particular phenotype[12,13]. The 4 individuals for whom detailed phenotypic follow-up has been possible have all proven healthy at ages 52, 73, 77, and 79, with no neurological phenotypes nor family history thereof[12,13]. This is evidence that heterozygous knockout of PrP is tolerated in humans. Using these population genomic databases we can also quantify "constraint", which is the degree to which loss-of-function mutations occur less frequently than expected based on DNA mutation rates[14,15]. Highly "constrained" genes are those where nonsense and frameshift variants appear to be removed quickly from the population, resulting in few or no individuals with these mutations in genomic databases. In contrast, *PRNP* gene is not "constrained" at all — these N-terminal knockout variants are found in the population at exactly the frequency expected based on DNA mutation rates, suggesting that these variants do not result in any premature mortality or infertility[13]. The occurrence in 1 in 25,000 individuals corresponds to an allele frequency of 1 in 50,000 alleles. This number squared is 1 in 2.5 billion, thus, we would expect a homozygous knockout genotype to be found in just a handful of humans on Earth, except in cases of consanguinity. Fewer than 1 million individuals have been sequenced and included in genomic databases to date. Thus, the fact that we have not yet observed a homozygous knockout human is numerically expected and does not provide any evidence that this genotype would not be tolerated[13].

Use of GMP material

The pivotal toxicology studies with siRNA 2439-exNA were conducted with the same GMP batch (Lot No. [redacted]) of drug substance that will be used clinically.

Formulation of material for toxicology studies and clinical formulation

In order to best represent the clinical Drug Product, the GLP toxicology studies also used sodium chloride, phosphate buffer, and magnesium chloride as excipients. The excipients are all compendial.

The clinical formulation of Drug Product, as described in 3.2.P.3.2, is shown in Table 2.6.6.1-1.

Table 2.6.6.1-1. Clinical formulation of 2439-exNA Drug Product

ingredient	concentration (mg/mL)	molarity (mM)
2439-exNA (label claim)	10 mg/mL	0.40
2438-exNA + oligonucleotide impurities*	10.369 mg/mL	0.42
potassium dihydrogen phosphate	0.144 mg/mL	1.06
sodium chloride	9.058 mg/mL	155.00
sodium phosphate dibasic heptahydrate	0.796 mg/mL	2.97
magnesium chloride hexahydrate	2.114 mg/mL	10.40

*Our Drug Substance was 96.44% pure by non-denaturing SEC (see section 3.2.S.4.4), therefore 10 mg/mL label claim means 10.369 mg/mL total oligonucleotide content. Oligonucleotide impurities will

also bind divalent cations, hence 10.369 mg/mL is the relevant figure for calculating the divalent cation ratio.

The clinical Drug Product formulation contains 10.40 mM of Mg^{2+} and 0.42 mM of drug plus oligonucleotide impurities, for a 25:1 molar ratio of divalent cation to oligonucleotide. It has been shown that a proper ratio of divalent cations, termed ionic conditioning, is needed to prevent neurotoxicity secondary to divalent cation imbalance[16–18]. Absent these cations, oligonucleotides can cause seizures believed to be due to high-affinity divalent cation binding sites on the nucleic acid backbone depleting calcium from CSF. The ratio of 25:1 was established as safe based on studies in both waking and anesthetized animals[16,18].

Accordingly, our toxicology studies also utilized a 25:1 molar ratio of divalent cation to oligonucleotide as shown in tables 2.6.6.1-2 and 2.6.6.1-3.

Table 2.6.6.1-2. Formulation of 2439-exNA for dog toxicology.

dose (mg) per animal	dose volume (mL)	drug + oligonucleotide impurities (mM)	magnesium chloride (mM)	sodium chloride (mM)	sodium phosphate (mM)	Mg^{2+}:oligo ratio
0	3	0.00	0.0	155.0	4.03	—
20	3	0.28	6.9	155.0	4.03	25.0
60	3	0.83	20.8	155.0	4.03	25.0
200	3	2.77	69.3	9.2	4.02	25.0

Table 2.6.6.1-3. Formulation of 2439-exNA for rat toxicology.

dose (mg) per animal	dose volume (mL)	drug + oligonucleotide impurities (mM)	magnesium chloride (mM)	sodium chloride (mM)	sodium phosphate (mM)	Mg^{2+}:oligo ratio
0	0.03	0.00	0.0	155.0	4.03	—
0.3	0.03	0.42	10.4	154.8	4.03	25.0
1.0	0.03	1.39	34.6	110.7	4.03	25.0
3.0	0.03	4.16	103.9	0.0	4.03	25.0

Our toxicology studies maintained the same 25:1 molar ratio of magnesium ions to siRNA as in the clinical formulation. As a result of this 25:1 ratio, a 200 mg dose of our clinical Drug Product in 20 mL volume contains 208 mmol of Mg^{2+}, the equivalent of 19.8 mg of magnesium chloride hexahydrate. This exceeds the quantity of intrathecal magnesium precedented in an FDA-approved drug according to the Inactive Ingredients Database, but its safety is supported by our toxicology studies, which also utilized this 25:1 magnesium:drug ratio. In dog toxicology studies, for instance, the maximum dose of 200 mg likewise contained 19.8 mg of magnesium chloride, in a species with only 12% the CSF volume of humans[19].

Toxicology studies conducted

A complete list of the nonclinical safety studies conducted, together with their GLP status, is included in (Table 1). All pivotal nonclinical safety studies were conducted in accordance with OECD (Organization for Economic Co-operation and Development) Test Guidelines and GLP. No adverse toxicity was identified in the completed 2439-exNA GLP toxicology studies in rats and dogs and 2439-exNA was not the genotoxic. This package of non-clinical toxicology studies is consistent with ICH M3 guidance and the completed animal studies supports the safety of the proposed clinical investigation with 2439-exNA in patients with or genetically at high risk for prion disease.

Table 2.6.6.2-1 Toxicology studies with 2439-exNA

Study type and duration	Dose/Concentration	Dosing Route	Species	Study Number	GLP compliant
Single dose toxicity:					
Single-Dose DRF toxicity study	200 mg	IT	Dog	A-703-03	No
Single-Dose pivotal toxicity study	0, 20, 60, and 200 mg	IT	Dog	A-703-02	Yes
Single-Dose pivotal toxicity study	0.3, 1.0, and 3.0 mg	IT	Rat	A-703-01	Yes
Genetic toxicology:					

Study type and duration	Dose/Concentration	Dosing Route	Species	Study Number	GLP compliant
Bacterial reverse mutation	0, 5.0, 16, 50, 160, 500, 1600, and 5000 µg/plate	In vitro	*Salmonella typhimurium* (TA1535, TA1537, TA98, and TA100) and *Escherichia coli* (WP2 trp uvrA) strains	56772	Yes
In vitro micronucleus assays	0, 7.5, 15, 30, 60, 1225, 250 and 500 µg/mL	In vitro	CHO-K1	56773	Yes

2.6.6.2 Single-Dose Toxicity

2.6.6.2.1 Dog

2.6.6.2.1.1 Maximum Tolerated Intrathecal Dose Study in Dogs (non-GLP; Study A-703-03)

The purpose of this study was to determine the maximum tolerated dose of 2439-exNA administered once via intrathecal injection to Beagle dogs. The highest dose was selected to be 200 mg in dog (16 mL CSF), in order to provide an approximately 8-fold margin, based on CSF volume scaling[19], over the max feasible clinical dose of 200 mg in human (130 mL CSF). One dog per sex was administered 200 mg 2439-exNA via intrathecal slow bolus over a duration of 5 minutes and observed for 4 days following dose administration. There were no clinical observations attributed to 2439-exNA following administration of the high dose of 200 mg. With no adverse effects following administration at 200 mg, the maximum feasible dose, additional dose levels were not evaluated. A dose of 200 mg 2439-exNA was considered well tolerated in Beagle dogs.

2.6.6.2.1.2 Single Intrathecal Dose Toxicology and Toxicokinetic Study in Dogs with a 28-Day Recovery Period (GLP; Study A-703-02)

The purpose of this study was to evaluate the toxicity, toxicokinetics, and the reversibility of any toxicological effects of 2439-exNA up to 200 mg when administered via intrathecal injection to beagle dogs.

Forty-eight (24 males/24 females) Beagle dogs were assigned to one of four dose groups: Group 1, Vehicle Control (0 mg), Group 2, Low (20 mg), Group 3, Mid (60 mg), and Group 4, High (200 mg) 2439-exNA. On Day 1, the animals were anesthetized and administered the appropriate test or control material via intrathecal injection administered over 3-5 minutes. On Day 2 four animals per sex per group and on Day 29 two animals per sex per group were humanely terminated and a necropsy with select tissue collection was performed.

There was no mortality observed. Clinical observations observed post dose administration on Day 1 consisted of decreased activity, lethargy, excessive salivation, impaired limb use/limb weakness. These observations resolved by Day 2 or 3 and were attributed to the anesthesia and intrathecal administration procedure. There were no 2439-exNA related effects on body weights, food consumption, neurobehavioral observations, ophthalmic observations, respiratory rates, ECG waveforms, or cardiovascular parameters. Additionally, there were no changes in hematology, coagulation, serum chemistry, and urine attributed to the administration of 2439-exNA.

T_{max} occurred at the first or second target sample collection time of 0.5 or 4.00 hr. There were no sex effects on the TK parameters of 2439-exNA. Systemic exposure increased in a greater than dose proportional manner between 20 to 60 mg, and with a near dose proportional increase from 60 to 200 mg.

The highest concentration of 2439-exNA was present on Day 2 in the liver. 2439-exNA was observed in the lumbar and cervical spinal cord, with higher concentrations observed in the lumbar spinal cord which correlates to the site of administration. siRNA 2439-exNA was observed in the brain, with frontal cortex, hippocampus, and cerebellar cortex generally containing higher concentrations than the thalamus, putamen, and caudate. 2439-exNA was also observed in the kidney. Generally, 2439-exNA remained present in liver, kidney, spinal cord, and brain at Day 29, but at lower concentrations compared to Day 2.

Microscopic test article related findings were limited to minimal focal to multifocal gliosis in the brain and spinal cord of one 60 mg 2439-exNA and four 200 mg 2439-exNA Core animals. Gliosis involved increases in indeterminate glial cells and therefore was non-specific evidence of inflammation or cell damage. Gliosis can manifest within 24 hours of injury and may remain for days or weeks, depending on the severity and chronicity of the inciting insult[20]. Three of the affected animals had minimal gliosis observed in the spinal cord at either the injection site (Animals 401 and 402) or in the thoracolumbar region (Animal 451), and two animals had the finding in the cerebrum (Animals 304 and 404). For Animal 404, the described focal gliosis was associated with focal necrosis secondary to injection in the spinal cord and therefore may be due only to this local cell damage and be independent of test article. For the remaining four animals, there was no indication of parenchymal injury (e.g., degeneration, necrosis, or inflammation) to incite gliosis. Additionally, these findings were not associated with any relevant clinical observations or neurobehavioral findings in any animals. There was no gliosis noted in recovery animals. The totality of data indicates that the gliosis was attributed to 2439-exNA administration and considered not adverse; consistent with published literature for this class of therapeutics[21,22].

In conclusion, there were no test article related adverse findings at 20 mg, and there were no adverse test article related findings at the 60 and 200 mg 2439-exNA dose levels. Non-adverse test article related findings in the 60 and 200 mg 2439-exNA groups were limited to gliosis. Therefore, 200 mg 2439-exNA is considered the no observable adverse effect level (NOAEL).

2.6.6.2.2 Rat

2.6.6.2.2.1 A Single Intrathecal Dose Toxicology and Toxicokinetic Study in Rats with a 28-Day Recovery Period (GLP; Study A-703-03)

The purpose of this study was to evaluate the potential toxicity and toxicokinetics of 2439-exNA following a single intrathecal injection to Sprague Dawley rats. The reversibility of any toxicological effects were evaluated following a 28-day recovery period.

One hundred sixty-two (81 males/81 females) Sprague Dawley rats were assigned to one of four dose groups: Group 1, Vehicle Control (0 mg), Group 2-Low (0.3 mg), Group 3-Mid (1.0 mg), and Group 4-High (3.0 mg) 2439-exNA. On Day 1, the animals were administered the appropriate test or control article via intrathecal bolus injection. On Days 2 and 29, core and recovery animals, respectively, were humanely terminated and a necropsy with select tissue collection was performed.

The formulation analysis resulted in average concentrations that were outside of the acceptance criteria with results of 116.4 to 122.7% of the nominal concentrations 2439-exNA. Based on the analysis performance, the reported results are considered representative of the concentrations of in the formulation samples, with the exceptions that the study samples are not within proven stability. No indication of sample degradation has been observed and most samples missed the acceptance criteria, showing that the nominal concentrations were exceeded in all analyte-containing formulations.

There were no clinical observations attributed to the administration of 2439-exNA. In addition, there were no changes in body weights, body weight gains, or food consumption attributed to the administration of 2439-exNA.

There were no ocular observations attributed to the intrathecal administration of 2439-exNA in Sprague Dawley rats.

TK parameters were determined and evaluated for 2439-exNA in male and female Sprague Dawley rats following intrathecal administration on Day 1. All control (Group 1) samples were BLOQ. All groups were reanalyzed without concentrations at 48 and 72 hr, except the Group 2 females which BLOQ at those time points, as these concentrations often increased from earlier time points and created apparent biphasic declines that may not be true characterizations of the elimination of 2439-exNA.

Following the dose administration, T_{max} occurred at the first target sample collection time of 0.5 hour. There were no apparent sex or dose effects on the TK parameters of siRNA 2439-exNA. Systemic exposure increased in a near dose proportional manner for both males and females as the dose was increased from 0.367 to 3.49 mg.

Tissue results are pending.

There were no changes in hematology, coagulation, serum chemistry, and urine parameters attributed to the administration of 2439-exNA. Sporadic statistically significant changes in some parameters relative to same-day Group 1 controls were either not dose-dependent limited in scope and considered due to normal biological variation.

Sprague Dawley rats given up to 3.0 mg 2439-exNA by intrathecal bolus injection on Day 1 and terminated on Day 2 had no increased mortality, no gross findings and no microscopic findings attributed to the administration of 2439-exNA. Similarly, recovery animals terminated on Day 29 had no findings attributed to the administration of 2439-exNA.

One Group 1 core female, one Group 2 core male, and one Group 4 core female had minimal to mild necrosis of deep skeletal muscle of the skin at the injection site. Two of these lesions were perpendicular to the skin surface, suggesting traumatic needle injury from the injection.

Overall, there were no observed adverse effects related to intrathecal administration of 2439-exNA to Sprague Dawley rats across the dose range tested of 0.3 to 3.0 mg. The No Observed Adverse Effect Level (NOAEL) was determined to be 3.0 mg.

2.6.6.4 Genotoxicity

2439-exNA was evaluated for genotoxic potential in alignment with Option 1 of ICH S2 (R1) (2011). The studies conducted were the in vitro bacterial reverse mutation assay (GLP) and the *vitro* micronucleus test with the Chinese hamster ovary derived CHO-K1 cells (GLP). The highest concentrations of 2439-exNA used for the in vitro assays, 5000 mcg/plate, were the maximum dose level recommended per ICH S2A(R1).

2.6.6.4.1 In vitro studies

2.6.6.4.1.1 Bacterial reverse mutation assay (GLP Study 56772)

The objective of the study was to evaluate the ability of 2439-exNA to induce reverse mutations in *Salmonella* and *E. coli* strains in the presence or absence of an exogenous mammalian metabolic activation system containing microsomal enzymes (S9). The study was conducted according to the ICH S2 (R1) and OECD-471 test guidelines.

Assay was performed using *Salmonella typhimurium* (TA1535, TA1537, TA98, and TA100) and *Escherichia coli* (WP2 trp uvrA) strains in the presence (+S9) and absence (-S9) of metabolic activation (rat liver homogenate fraction). Seven dose levels of 2439-exNA (5.0, 16, 50, 160, 500, 1600 and 5000 µg/plate) along with the concurrent positive and vehicle/negative controls were tested for the mutation assay.

The revertant colony numbers obtained from the vehicle/negative and positive controls in all the five tester strains (TA1535, TA1537, TA98, TA100, and WP2 trp uvrA), met the study plan acceptance criteria. The mean revertant counts obtained from the five bacterial strains treated with SWFI (vehicle/negative control used for 2439-exNA and Vehicle 1 used for SAZ), and DMSO (Vehicle 2 for 2NF, 9AC, NQO, 2AA, and B[a]P) were within the limits of the laboratory's historical control range. The mean revertant counts expressed by the positive controls were within the established laboratory's historical control data range. The positive control items, with or without S9, produced the expected increase in revertant colony counts to at least twice the concurrent vehicle/negative control levels, confirming the sensitivity and validity of the test system. Based on these results, the assay was considered to have met the required criteria for a valid assay in all strains.

No 2439-exNA precipitates were observed, and there was no thinning of the background lawn or significant reduction in revertant colony counts (< 0.5-fold) compared to vehicle/negative control observed in any doses or conditions tested. Therefore, 2439-exNA was not considered cytotoxic to the bacterial system.

2439-exNA did not induce an increase in the revertant frequencies (≥ 2-fold) as compared to that of the vehicle/negative controls, for all concentrations tested under both non-activated as well as S9-activated metabolic conditions.

In conclusion, under the conditions of this study, the test item 2439-exNA was devoid of any mutagenic potential in the bacterial reverse mutation test (Ames test), when tested on *S. typhimurium* strains TA1535, TA1537, TA98, TA100, and on the *E. coli* strain WP2 trp uvrA, with and without metabolic activation (±S9), at dose levels up to the Guideline limit dose of 5000 µg/plate.

2.6.6.4.1.2 Micronucleus assay (GLP Study 56773)

The objective of this study was to evaluate the cytogenetic potential of the test item, 2439-exNA, using the *in vitro* micronucleus test with the Chinese hamster ovary derived CHO-K1 cells. The study was conducted according to the ICH S2 (R1) and OECD-487 test guidelines.

Exponentially growing CHO-K1 cells were treated in duplicate with 2439-exNA at a range of concentrations up to the highest dose of 500 µg/mL in the presence (4-hour treatment) or absence (4-hour and 26-hour treatments) of metabolic activation (rat liver S9 microsomal fraction [S9]). Concurrent vehicle/negative and positive control items were included for each treatment regimen. Cells were evaluated for cytotoxicity and micronuclei count by flow cytometry (~20,000 nuclei per dose).

The degree of cytotoxicity following treatment was evaluated by determining the Relative Population Doubling (RPD). As recommended by the ICH and OECD-487 test guidelines, the highest concentrations chosen for analysis of micronucleus aimed to achieve cytotoxicity of approximately 55 ± 5% (i.e., reduction in RPD to 45 ± 5% compared to concurrent vehicle/negative control). No 2439-exNA precipitate was observed (checked by visual and microscopic examination) at the beginning and end of the treatment in all 3 regimens.

2439-exNA did not show any significant decrease (i.e., <45 ± 5%) in mean RPD at any tested concentration and conditions, indicating that 2439-exNA was non-cytotoxic to the test system in the absence or presence of S9. Therefore, the highest tested concentration of 500 µg/mL (standard limit dose) was chosen for the micronucleus evaluation. Two additional lower concentrations (250 and 125 µg/mL) were also analyzed for micronucleus evaluation in each of the 3 treatment regimens.

The results obtained with vehicle/negative control groups fell within or close to the 95% confidence limits (CI) of the distribution of the laboratory historical vehicle/negative control data range. At least one dose levels of the positive control induced a statistically significant increase in micronuclei or hypodiploid frequencies ($p \leq 0.01$). The results obtained from concurrent positive and vehicle/negative controls confirmed the sensitivity of the test system, the effectiveness of the S9 mix, and the validity of the assay.

Cultures treated with 2439-exNA did not show any statistically significant increase in the incidence of micronuclei. The numbers of micronuclei were comparable with the concurrent vehicle/negative control and were within or close to the historical control range. There was no significant increase in hypodiploid cells, indicative of numerical aberrations, in either the short-duration or extended treatments with 2439-exNA, in the absence or presence of S9.

In conclusion, 2439-exNA was considered negative for clastogenic or aneugenic activity in the *in vitro* micronucleus assay based on the lack of increased micronuclei or hypodiploid frequencies in CHO-K1 cells following 4-hour incubation with and without metabolic activation and 26-hour incubation without metabolic activation at dose levels up to the Guideline limit dose of 500 µg/mL

2.6.6.5 Carcinogenicity

No carcinogenicity studies have been performed.

2.6.6.6 Reproductive and Developmental Toxicity

No reproductive and developmental toxicity studies have been performed.

2.6.6.7 Local Tolerance

No local tolerance studies have been performed.

2.6.6.8 Other Toxicity Studies

No other toxicity studies have been performed.

2.6.6.9 Discussion and Conclusion

The toxicology studies conducted with 2439-exNA adequately characterized the nonclinical safety of 2439-exNA.

Single dose GLP nonclinical toxicology studies were conducted in two species, the rat and dog. The studies used intrathecal administration, which is the intended route of administration for patients. The toxicology assessment did not identify adverse toxicity following a single IT dose of 2439-exNA of up to 3.0 mg in rat and 200 mg in the dog GLP toxicology studies.

2439-exNA was not genotoxic; 2439-exNA was negative in both the bacterial reverse mutation assay (i.e., non-mutagenic) and in the chromosomal aberration assay in cultured Chinese hamster lung fibroblast with or without metabolic activation (i.e., non-clastogenic).

Based on the nonclinical safety data, the proposed clinical starting dose is 50 mg, which provides a minimum >26-fold margin when calculated by local concentration at the injection site or by CSF volume scaling (Table 2.6.6.9-1) . The highest proposed clinical dose, 200 mg, has a margin of 6-10-fold.

Table 2.6.6.9-1. Proposed dose levels compared to toxicology data.

species	CSF volume (mL)	dose volume (mL)	NOAEL (mg)	proposed human doses (mg)	local concentration (mg/mL)	CSF concentration (mg/mL)	margin by local vs. rat	margin by local vs. dog	margin by CSF vs. rat	margin by CSF vs. dog
rat	0.25	0.03	3	–	100.0	12.0				
dog	16	3	200	–	66.7	12.5				
human	130	20	–	50	2.5	0.4	40.0	26.7	31.2	32.5
				100	5.0	0.8	20.0	13.3	15.6	16.3
				200	10.0	1.5	10.0	6.7	7.8	8.1

Safety margins to the proposed dose levels in humans to the NOAELs for rat and dog, using local injection concentration and CSF volumes (Emami et al[19,23])

2.6.6.10 Literature References

1. Vallabh SM, Minikel EV, Schreiber SL, Lander ES. Towards a treatment for genetic prion disease: trials and biomarkers. Lancet Neurol. 2020 Apr;19(4):361–368. PMID: 32199098

2. Büeler H, Fischer M, Lang Y, Bluethmann H, Lipp HP, DeArmond SJ, Prusiner SB, Aguet M, Weissmann C. Normal development and behaviour of mice lacking the neuronal cell-surface PrP protein. Nature. 1992 Apr 16;356(6370):577–582. PMID: 1373228

3. Nuvolone M, Hermann M, Sorce S, Russo G, Tiberi C, Schwarz P, Minikel E, Sanoudou D, Pelczar P, Aguzzi A. Strictly co-isogenic C57BL/6J-Prnp-/- mice: A rigorous resource for prion science. J Exp Med. 2016 Mar 7;213(3):313–327. PMCID: PMC4813672

4. Richt JA, Kasinathan P, Hamir AN, Castilla J, Sathiyaseelan T, Vargas F, Sathiyaseelan J, Wu H, Matsushita H, Koster J, Kato S, Ishida I, Soto C, Robl JM, Kuroiwa Y. Production of cattle lacking prion protein. Nat Biotechnol. 2007 Jan;25(1):132–138. PMCID: PMC2813193

5. Yu G, Chen J, Xu Y, Zhu C, Yu H, Liu S, Sha H, Chen J, Xu X, Wu Y, Zhang A, Ma J, Cheng G. Generation of goats lacking prion protein. Mol Reprod Dev. 2009 Jan;76(1):3. PMID: 18951376

6. Benestad SL, Austbø L, Tranulis MA, Espenes A, Olsaker I. Healthy goats naturally devoid of prion protein. Vet Res. 2012;43:87. PMCID: PMC3542104

7. Mallucci GR, Ratté S, Asante EA, Linehan J, Gowland I, Jefferys JGR, Collinge J. Post-natal knockout of prion protein alters hippocampal CA1 properties, but does not result in neurodegeneration. EMBO J. 2002 Feb 1;21(3):202–210. PMCID: PMC125833

8. Wulf MA, Senatore A, Aguzzi A. The biological function of the cellular prion protein: an update. BMC Biol. 2017 02;15(1):34. PMCID: PMC5412054

9. Bremer J, Baumann F, Tiberi C, Wessig C, Fischer H, Schwarz P, Steele AD, Toyka KV, Nave KA, Weis J, Aguzzi A. Axonal prion protein is required for peripheral myelin maintenance. Nat Neurosci. 2010 Mar;13(3):310–318. PMID: 20098419

10. Skedsmo FS, Malachin G, Våge DI, Hammervold MM, Salvesen Ø, Ersdal C, Ranheim B, Stafsnes MH, Bartosova Z, Bruheim P, Jäderlund KH, Matiasek K, Espenes A, Tranulis MA. Demyelinating polyneuropathy in goats lacking prion protein. FASEB J. 2019 Dec 13; PMID: 31907995

11. Küffer A, Lakkaraju AKK, Mogha A, Petersen SC, Airich K, Doucerain C, Marpakwar R, Bakirci P, Senatore A, Monnard A, Schiavi C, Nuvolone M, Grosshans B, Hornemann S, Bassilana F, Monk KR, Aguzzi A. The prion protein is an agonistic ligand of the G protein-coupled receptor Adgrg6. Nature. 2016 Aug 25;536(7617):464–468. PMID: 27501152

12. Minikel EV, Vallabh SM, Lek M, Estrada K, Samocha KE, Sathirapongsasuti JF, McLean CY, Tung JY, Yu LPC, Gambetti P, Blevins J, Zhang S, Cohen Y, Chen W, Yamada M, Hamaguchi T, Sanjo N, Mizusawa H, Nakamura Y, Kitamoto T, Collins SJ, Boyd A, Will RG, Knight R, Ponto C, Zerr I, Kraus TFJ, Eigenbrod S, Giese A, Calero M, de Pedro-Cuesta J, Haïk S, Laplanche JL, Bouaziz-Amar E, Brandel JP, Capellari S, Parchi P, Poleggi A, Ladogana A, O'Donnell-Luria AH, Karczewski KJ, Marshall JL, Boehnke M, Laakso M, Mohlke KL, Kähler A, Chambert K, McCarroll S, Sullivan PF, Hultman CM, Purcell SM, Sklar P, van der Lee SJ, Rozemuller A, Jansen C, Hofman A, Kraaij R, van Rooij JGJ, Ikram MA, Uitterlinden AG, van Duijn CM, Exome Aggregation Consortium (ExAC), Daly MJ, MacArthur DG. Quantifying prion disease penetrance using large population control cohorts. Sci Transl Med. 2016 Jan 20;8(322):322ra9. PMCID: PMC4774245

13. Minikel EV, Karczewski KJ, Martin HC, Cummings BB, Whiffin N, Rhodes D, Alföldi J, Trembath RC, van Heel DA, Daly MJ, Genome Aggregation Database Production Team, Genome Aggregation Database

Consortium, Schreiber SL, MacArthur DG. Evaluating drug targets through human loss-of-function genetic variation. Nature. 2020 May;581(7809):459–464. PMCID: PMC7272226

14. Samocha KE, Robinson EB, Sanders SJ, Stevens C, Sabo A, McGrath LM, Kosmicki JA, Rehnström K, Mallick S, Kirby A, Wall DP, MacArthur DG, Gabriel SB, DePristo M, Purcell SM, Palotie A, Boerwinkle E, Buxbaum JD, Cook EH, Gibbs RA, Schellenberg GD, Sutcliffe JS, Devlin B, Roeder K, Neale BM, Daly MJ. A framework for the interpretation of de novo mutation in human disease. Nat Genet. 2014 Sep;46(9):944–950. PMCID: PMC4222185

15. Karczewski KJ, Francioli LC, Tiao G, Cummings BB, Alföldi J, Wang Q, Collins RL, Laricchia KM, Ganna A, Birnbaum DP, Gauthier LD, Brand H, Solomonson M, Watts NA, Rhodes D, Singer-Berk M, England EM, Seaby EG, Kosmicki JA, Walters RK, Tashman K, Farjoun Y, Banks E, Poterba T, Wang A, Seed C, Whiffin N, Chong JX, Samocha KE, Pierce-Hoffman E, Zappala Z, O'Donnell-Luria AH, Minikel EV, Weisburd B, Lek M, Ware JS, Vittal C, Armean IM, Bergelson L, Cibulskis K, Connolly KM, Covarrubias M, Donnelly S, Ferriera S, Gabriel S, Gentry J, Gupta N, Jeandet T, Kaplan D, Llanwarne C, Munshi R, Novod S, Petrillo N, Roazen D, Ruano-Rubio V, Saltzman A, Schleicher M, Soto J, Tibbetts K, Tolonen C, Wade G, Talkowski ME, Genome Aggregation Database Consortium, Neale BM, Daly MJ, MacArthur DG. The mutational constraint spectrum quantified from variation in 141,456 humans. Nature. 2020;581(7809):434–443. PMCID: PMC7334197

16. Miller R, Paquette J, Barker A, Sapp E, McHugh N, Bramato B, Yamada N, Alterman J, Echeveria D, Yamada K, Watts J, Anaclet C, DiFiglia M, Khvorova A, Aronin N. Preventing acute neurotoxicity of CNS therapeutic oligonucleotides with the addition of Ca2+ and Mg2+ in the formulation. Mol Ther Nucleic Acids. 2024 Dec 10;35(4):102359. PMCID: PMC11567125

17. Moazami MP, Rembetsy-Brown JM, Sarli SL, McEachern HR, Wang F, Ohara M, Wagh A, Kelly K, Krishnamurthy PM, Weiss A, Marosfoi M, King RM, Motwani M, Gray-Edwards H, Fitzgerald KA, Brown RH, Watts JK. Quantifying and mitigating motor phenotypes induced by antisense oligonucleotides in the central nervous system. Mol Ther. 2024 Oct 28;S1525-0016(24)00684–1. PMID: 39460376

18. Hassler M, Kinberger GA, Godinho BM da C. Compositions and methods for delivering therapeutic oligonucleotides to the central nervous system [Internet]. US20240287526A1, 2024 [cited 2024 Dec 27]. Available from: https://patents.google.com/patent/US20240287526A1/en

19. Emami A, Tepper J, Short B, Yaksh TL, Bendele AM, Ramani T, Cisternas AF, Chang JH, Mellon RD. Toxicology Evaluation of Drugs Administered via Uncommon Routes: Intranasal, Intraocular, Intrathecal/Intraspinal, and Intra-Articular. Int J Toxicol. 2018;37(1):4–27. PMCID: PMC5874330

20. Burda JE, Sofroniew MV. Reactive gliosis and the multicellular response to CNS damage and disease. Neuron. 2014 Jan 22;81(2):229–248. PMCID: PMC3984950

21. Berman-Booty LD, Klein SK, Mazur C, Schroeder J, Korte S, Ludwig FT, Romeike A, Bolon B, Grieves JL. Toxicologic Pathology Forum: Opinion on Interpretive Challenges for Procedure-Related Effects Associated With Direct Central Nervous System Delivery of Oligonucleotides to Rodents, Dogs, and Nonhuman Primates. Toxicol Pathol. 2023 Aug;51(6):375–389. PMID: 38179962

22. Bolon B. Toxicologic Pathology Forum Opinion: Interpretation of Gliosis in the Brain and Spinal Cord Observed During Nonclinical Safety Studies. Toxicol Pathol. 2023 Jan;51(1–2):68–76. PMID: 37057409

23. Emami A, Chang JH, Mellon D, Yaksh TL. Local safety assessment of drugs administered via the intrathecal route. Poster. American College of Toxicology 44th Annual Meeting. International Journal of Toxicology [Internet]. 2023. p. 73–115. Available from: https://doi.org/10.1177/10915818231220242